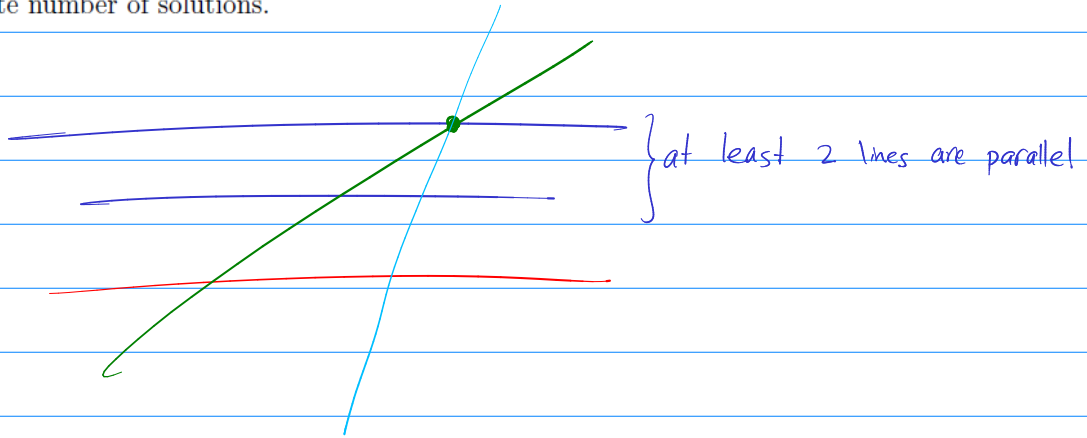


Week 1

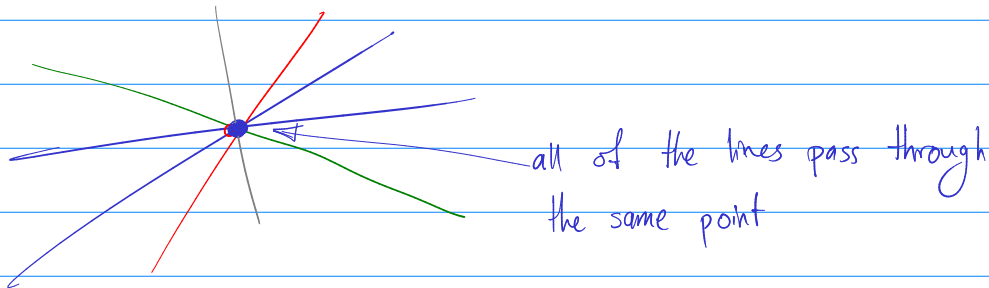
You have a system of k equations in two variables, $k \geq 2$. Explain the geometric significance of

- (a) No solution.
- (b) A unique solution.
- (c) An infinite number of solutions.

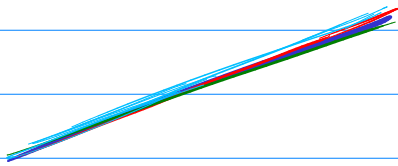
a)



b)



c)



all of the lines are essentially the same.

Here is an augmented matrix in which * denotes an arbitrary number and ■ denotes a nonzero number. Determine whether the given augmented matrix is consistent. If consistent, is the solution unique?

corresponding system

$$\left(\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & * & * & 0 & * \\ 0 & 0 & \blacksquare & * & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * \end{array} \right)$$

each row has a leading 1 (full row rank) $\Rightarrow$ Consistent

A column with no leading 1 (free variable) $\Rightarrow$ infinitely many solutions

Here is an augmented matrix in which * denotes an arbitrary number and ■ denotes a nonzero number. Determine whether the ~~given augmented matrix~~ ^{corresponding system} is consistent. If consistent, is the solution unique?

$$\left(\begin{array}{ccc|c} \blacksquare & * & * & * \\ 0 & \blacksquare & * & * \\ 0 & 0 & \blacksquare & * \end{array} \right)$$

Each row has a leading 1 (full row rank) $\Rightarrow$ consistent
Each column has a leading 1 (no free variables) $\Rightarrow$ unique solution

Here is an augmented matrix in which * denotes an arbitrary number and ■ denotes a nonzero number. Determine whether the given augmented matrix is consistent. If consistent, is the solution unique?

$$\left(\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & 0 & * & 0 & * \\ 0 & 0 & 0 & \blacksquare & * & * \\ 0 & 0 & 0 & 0 & \blacksquare & * \end{array} \right)$$

Each row has a leading 1 (full row rank) $\Rightarrow$ Consistent
There is a column with no leading 1 (free variable) $\Rightarrow$ infinitely many solutions

Here is an augmented matrix in which * denotes an arbitrary number and ■ denotes a nonzero number. Determine whether the given augmented matrix is consistent. If consistent, is the solution unique?

$$\left(\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & * & * & 0 & * \\ 0 & 0 & 0 & 0 & \blacksquare & 0 \\ 0 & 0 & 0 & 0 & * & \blacksquare \end{array} \right)$$

The next step in Gauss-Jordan elimination will make this entry zero and this entry remains nonzero (because of this zero):

$$\left(\begin{array}{ccccc|c} \blacksquare & * & * & * & * & * \\ 0 & \blacksquare & * & * & 0 & * \\ 0 & 0 & 0 & 0 & \blacksquare & 0 \\ 0 & 0 & 0 & 0 & 0 & \blacksquare \end{array} \right)$$

$$\rightarrow 0 \cdot x + 0 \cdot y + 0 \cdot z + 0 \cdot v + 0 \cdot w = \blacksquare$$

$$\text{i.e. } 0 = \blacksquare$$

↖ a nonzero number

⇒ The system doesn't have a solution. i.e. it is inconsistent.

Choose h and k such that the augmented matrix shown has one solution. Then choose h and k such that the system has no solutions. Finally, choose h and k such that the system has infinitely many solutions.

$$\left(\begin{array}{cc|c} 1 & h & 2 \\ 2 & 4 & k \end{array} \right).$$

↓ Gauss-Jordan elimination

leading 1

$$\left(\begin{array}{cc|c} 1 & h & 2 \\ 0 & 4-2h & k-4 \end{array} \right)$$

↓ depends if this is zero or not

→ { If $4-2h \neq 0 \Rightarrow h \neq 2$ then there is a ^{unique} solution.

{ If $4-2h = 0 \Rightarrow h = 2 \rightarrow$ { If $k = 4 : \left(\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & 0 & 0 \end{array} \right)$
then there are infinitely many solutions.

{ If $k \neq 4 : \left(\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & 0 & \blacksquare \end{array} \right)$
no solutions.

Find the general solution of the system whose augmented matrix is

leading 1 $\rightarrow \begin{pmatrix} 1 & 1 & 0 & | & 1 \\ 1 & 0 & 4 & | & 2 \end{pmatrix}$.

$R_2 - R_1 \rightarrow \begin{pmatrix} 1 & 1 & 0 & | & 1 \\ 0 & -1 & 4 & | & -1 \end{pmatrix} \xrightarrow{R_2 \cdot (-1)} \begin{pmatrix} 1 & 1 & 0 & | & 1 \\ 0 & 1 & -4 & | & -1 \end{pmatrix}$

$R_1 - R_2 \rightarrow \begin{pmatrix} 1 & 0 & 4 & | & 2 \\ 0 & 1 & -4 & | & -1 \end{pmatrix}$

$x \ y \ z$

$x + 4z = 2 \Rightarrow x = -4s + 2$

$y - 4z = -1 \Rightarrow y = 4s - 1$

free variable: $z = s$, $s \in \mathbb{R}$

General solution: $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -4s + 2 \\ 4s - 1 \\ s \end{pmatrix}$, for $s \in \mathbb{R}$.

Give the complete solution to the system of equations, $3x - y + 4z = 6$, $y + 8z = 0$, and $-2x + y = -4$.

$$\begin{bmatrix} 3 & -1 & 4 & | & 6 \\ 0 & 1 & 8 & | & 0 \\ -2 & 1 & 0 & | & -4 \end{bmatrix} \xrightarrow{\text{ref}} \begin{bmatrix} \overset{x}{1} & \overset{y}{0} & \overset{z}{4} & | & 2 \\ 0 & 1 & 8 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

free variable: $z=t$, for $t \in \mathbb{R}$

$$\text{General solution: } \begin{cases} x = 2 - 4t \\ y = -8t \\ z = t \end{cases}, \text{ for } t \in \mathbb{R}.$$